

**MAHARASHTRA STATE
BOARD OF TECHNICAL EDUCATION, MUMBAI**

SEMESTER VI PROJECT PROPOSAL

**Hostel Attendance System Based on
Finger Print Authentication**

SUBMITTED BY

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UNDER THE GUIDANCE OF

Prof. Nayana C. Borase

SUBMITTED TO

DIPLOMA IN COMPUTER ENGINEERING

S.M.D.R. GOVERNMENT POLYTECHNIC DHULE

Year 2023-24



S.M.D.R. GOVERNMENT POLYTECHNIC DHULE

Project on
**Hostel Attendance System Based on
Finger Print Authentication**

A PROJECT REPORT

GUIDED BY

Prof. N. C. Borse Mam

DIPLOMA IN ENGINEERING

(Computer Engineering)

SUBMITTED TO

Maharashtra State Board of Technical Education, Mumbai

ACKNOWLEDGEMENT

The success and final outcome of this project required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of our project. All that we have done is only due to such supervision and assistance and we would not forget to thank them.

We respect and thank Dr. Mr. R. G. Wadekar, Principal, S.M.D.R. Government Polytechnic, Dhule for providing us an opportunity to do the project work in the institute and giving us all support and guidance which made us complete the project duly. We are extremely thankful to Mr. Wadekar Sir for providing such a nice support and guidance.

We heartily thank Dr. Mr. U. R. Deshmukh, Head, Department of Computer Engineering for his guidance, suggestions and encouragement during this project work.

We own our deep gratitude to our project guide Prof.Mrs. N.C.Borse, who took keep interest on our project work and guided us all along, till the completion of our project work by providing all the necessary information for developing a good system.

We are thankful to and fortunate enough to get constant encouragement, support and guidance from all the Teaching staffs of Department of Science and Humanities which helped us in successfully completing our project work.



CERTIFICATE

I hereby certify that the work which is being presented in the Diploma in Engineering. Project Plan Proposal entitled “**Hostel Attendance System**”, in partial fulfillment of the requirements for the award of the **Diploma in Computer Engineering** and submitted to the Department of Computer Engineering of S.M.D.R. Government Polytechnic Dhule , is an authentic record of my own work carried out during a period from **Jan 2024 to April 2024 (6th semester)** under the Guidance of **Prof. N.C. Borse Mam , Computer Engineering Department.**

This is to certify that the above statement made by the students is correct to the best of my knowledge.

Signature of Guide

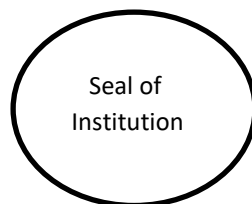
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**PROGRESSIVE ASSESSMENT (PA) OF CAPSTONE PROJECT-EXECUTION
AND REPORT WRITING**

Evaluation Sheet for Internal Assessment

Name of Student: Renuka,Unnati,Bhavana,Harshada

Name of Programme: Diploma in Computer Engineering

Semester: Sixth

Course Title: Capstone Project: Execution and Report Writing

Code:22060

Title of the Capstone Project: Hostel Attendance System Based on Finger Print Authentication

A. POs addressed by the capstone Project (Mention only those predominant POs)

- a)
- b)
- c)
- d)

B. COs addresses by the Capstone Project (Mention only those Predominant POs)

- a)
- b)
- c)
- d)

C. Other Learning outcomes achieved through this project

1. Unit Outcomes (Cognitive Domain)

- a)
- b)
- c)
- d)

2. Practical Outcomes (in psychomotor Domain)

- a)
- b)
- c)
- d)

3. Affective Domain Outcomes

- a)
- b)
- c)
- d)

PROGRESSIVE ASSESSEMENT (PA) SHEET			
Sr. No	Criteria	Max Marks	Obtained Marks
1	Project Proposal/Identification	10	
2	Punctuality and overall contribution		
3	Project Diary		
4	Execution of Plan during Sixth Semester	20	
5	Project report including documentation	10	
6	Presentation	05	
Total		50	

ABSTRACT

The biometric system plays the most important role in this current century. Finger print identification is one among the foremost distinguished and familiar identity verification system due to its individuality. Security within the hostel is one of the foremost repetitive issues. To keep up day by day attendance verification is sophisticated and time consuming system for the hostel management. There are number of existing attending systems are available for college students, for hostel students it must improve. Within the existing system wardens are manually maintain the attendance for hostel students. This paper deals with, avoid of an entire problem in hostel management system together with this monitoring system also proposed. The administrator of this system was college principal or warden. Biometric system is used to accommodate a large number of students within the hostel. This system makes automatically to monitor the entry and exit of students from hostel and offers alert SMS to parents for their safety.

Our project aims at designing an student attendance system which could effectively manage attendance of students at Hostel. Attendance is marked after student identification. For student identification, a fingerprint recognition based identification system is used. Fingerprints are considered to be the best and fastest method for biometric identification.

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Chapter 1

Introduction

Fingerprint identification is one of the most well-known and common biometric authentication systems. The system is specially proposed for hostel management and monitoring. Security in the organization is one of the most persistent issues that organization must address. Nowadays security is a prime concern in each individual's life. Currently student's attendance becomes more vital element for every organization. Entering and monitoring of hostel attendance is an area of administration that can involve essential of time and effort in a hostile atmosphere.

A biometrics system basically comprises entering and scrutiny biometric characteristics. When a person first uses a biometrics system, his or her characteristic options are entered as a reference for future comparison. Identification of the drawbacks in the present system leads to the designing of computerized system that may be compatible and user friendly.

Manual attendance taking and report generation has its limitations. It is well enough for 30-60 students but when it comes to taking attendance of students large in number, it is difficult. For taking attendance for a lecture, a hostel attendance, a conference, etc. roll calling and manual attendance system is a failure. Time waste over responses of students, waste of paper etc. are the disadvantages of manual attendance system. Moreover, the attendance report is also not generated on time. Attendance report which is circulated over NITR webmail is two months old. To overcome these non-optimal situations, it is necessary that we should use an automatic on-line attendance management system. So we present an implementable attendance management framework. Student attendance system framework is divided into three parts : Hardware/Software Design, Attendance Management Approach and On-line Report Generation. Each of these is explained below.

1.1 What is Biometric ?

Biometric Identification Systems are widely used for unique identification of humans mainly for verification and identification. Biometrics is used as a form of identity access management and access control. So use of biometrics in student attendance management system is a secure approach. There are many types of biometric systems like fingerprint recognition, face recognition, voice recognition, iris recognition, palm recognition etc. In this project, we used fingerprint recognition system.

Managing attendance records of students of an institute is a tedious task. It consumes time and paper both. To make all the attendance related work automatic and on-line, we have designed an attendance management system which could be implemented in NIT Rourkela. It uses a fingerprint identification system developed in this project. This fingerprint identification system uses existing as well as new techniques in fingerprint recognition and matching. A new one to many matching algorithm for large databases has been introduced in this identification system.

Fingerprint images that are found or scanned are not of optimum quality. So we remove noises and enhance their quality. We extract features like minutiae and others for matching. If

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the sets of minutiae are matched with those in the database, we call it an identified fingerprint. After matching, we perform post-matching steps which may include showing details of identified candidate, marking attendance etc. A brief flowchart is shown in next section.

This system is designed in favor of the hostel attendance which helps them to save the records of the students about their rooms and other things. It helps them from the manual work from which it is very difficult to find the record of the students and the information of about the those ones who had left the hostel . We design this system on the request of the hostel management, through this they cannot require so efficient person to handle and calculate the things.

This system automatically show the absent students records and issued the notifications for those students who are not taken permission to go out of hostel.

Chapter 2

Literature Review

The existing system largely consist two approaches for taking attendance of students. First is traditional approach, it consists of physical register where the supervisor manually inputs the attendance record of all students. Second approach is computing approach which has been developed to replace this manual system. It includes Desktop application, Web application, Bio-metric Attendance machine, Android application. The problem with existing system is that the manual system is time consuming and the advanced technologies come with their own shortcomings. Existing approaches for managing attendance are:

- Traditional Approach Step

Step 1: Manual Attendance Management System

- Computing Approach
 1. Biometric Attendance Machine
 2. Web Applications for Attendance Management
 3. Desktop Applications for Attendance Management
 4. Android Apps for Attendance Management

2.1. Manual Approach

Attendance is a way through which a teacher records the present and absent Students' list or data. In this approach a teacher records the data such as roll number and name of student; date etc. in a log book or a sheet (commonly known as attendance sheet). Attendance is taken through either by calling the roll number/name of student or by taking the signature of the student. Later, the data is entered in desktop application for example MS-Excel or MS-Word to prepare reports.

2.1.1. Drawbacks of Manual Approach

- Sheets may be lost or misplaced.
- Wrong data can be entered while taking attendance.
- This method include lots of manual work hence it is time consuming and inefficient.
- More use of Natural Resources (Paper).
- Since sheets or log books are use this increases the cost of this method.
- Inconsistency in data entry and generate errors.
- Lack of security.
- Report generation is difficult.

2.2. Biometric Attendance Machine

A Biometric attendance machine is a device that records the attendance of the student either by their fingerprint or by their EM card. When the student enters in the premises, he/she should use the Biometric attendance machine for their entry. Some Biometric attendance machine uses the EM card provided to student for the same job (this is mostly use in the IIT coaching institution). This is more secure way for the attendance management, as only the authorized person can retrieve data from this machine. Also it provides accurate data.

2.2.1. Drawbacks of Biometric Attendance Machine

- It is physically less protected.
- Its cost is too high.
- It is less flexible.
- Report cannot be easily generated as per need.
- It has limited amount of entries.
- It has fewer variations to perform different task.
- It becomes time consuming method when the number of students increases.
- Proxy can be easily done when EM card are used (but not in the case of fingerprint).

2.3. Web Applications for Attendance Management

This methodology contains three major phases - attendance marking, and attendance management and report generation. User authentication is one of the major factors in attendance monitoring system. Every student is authenticated based on his/her unique user identification number/code. Only the admin, the concerned staff or faculty member is given the authority to generate report. The report is generated by selecting 'from' and 'to' dates. We can view the subject code, subject name, total number of classes conducted, total number of classes attended and the attendance percentage. Separate modules are written for each of the functions.

2.3.1. Drawbacks of Web Application for Attendance Management

- It needs an Internet Connection, so if there is no availability of Network (Internet) we can't use it.
- It is not much user-friendly. User needs to learn "How to use it?" first, only then he can do his/her job on it.
- Every time user needs to log-in to take attendance.
- It is time consuming when user needs the data as per his/her need. Data is not available all the time, as user need to log-in every time to fetch his/her data.
- It needs the platforms/devices such as PC, Laptop or Smart Phones. PCs are not mobile devices and since they are cost to be made available at each place, so they are not efficient. Laptops and Smart Phones are mobile but they are worthless if Internet connection is slow or not available.

2.4. Desktop Applications for Attendance Management

There are only few applications that are developed particularly for the attendance management job. And from those, some of them are open source and some are not, such as "Zoho People", "Bitrix 24", "Avaza" company/organization. Some users think of taking attendance in MS-Excel, while other prefers to use other application. In such type of approach the attendance is taken through entering either present or absent roll number of the students. The report generation is much simpler than the previous systems. And it is secure since the data can be easily encrypted through password.

2.4.1. Drawbacks of the Desktop application for attendance management

- Desktop application attendance management runs on the PCs or Laptops. Laptops are mobile but for such a small job (for attendance), it is not good idea to carry it always. PCs are not mobile so they cannot be carried by user everywhere he/she needs it.
- Platforms for desktop applications are costly.
 - No backups are available when the system gets failed or crashed.
- Desktop applications for attendance management are less User-friendly.
- Desktop applications for attendance management less flexible.
- PCs requires power continuously, it does not have battery backups, so in case when the electricity is not available this become worthless.

2.5. Android Apps for Attendance Management

Android Apps are much flexible and provide more features than all previous systems. It is user friendly, requires less energy, and open source, for example - "Smart Attendance Manager" designed and developed by "Alienart softwares". In this app, to create a new attendance sheet we need to provide it the subject details [subject name, department name, division], type [lecture, practical, seminar, exam], including year/standard, student details, and schedule details. Attendance method is very easy as user needs to just tap on those roll numbers which are absent. Report generation is easy and much user-friendly.

2.5.1. Drawbacks of existing Android Apps for attendance management

- Backups are not present.
 - User can access the data on same device where he/she has stored it, i.e. where the app is installed and used. User cannot access the data on another device.
 - List of students roll numbers cannot be extended once entered.
 - Limitation of attendance percentage is not highlighted.
 - Another app is needed for printing the data.
- Weekly or monthly reports are not generated.

Chapter 3

Scope of Project

3.1 Problem Statement

Design a hostel attendance system using biometric fingerprint technology implemented in Python. The system should accurately record the attendance of hostel residents based on their individual fingerprints. Additionally, it should send a notification message to the respective student's parents or guardians to inform them about their child's attendance status.

1. Biometric Fingerprint Integration: Develop a module that can capture and verify the fingerprints of hostel residents.
2. Attendance Recording: Implement a mechanism to record the attendance of students based on their fingerprints when they enter or exit the hostel premises.
3. Parent Notification: Create a feature that sends an automated message to the registered mobile number of the student's parents or guardians, informing them about their child's attendance (in or out) and any other relevant details.
4. Data Security: Ensure that biometric data is securely stored and processed to protect the privacy and security of the residents.
5. Integration with Hostel Management System: Integrate the attendance system with the existing hostel management system to facilitate seamless data flow and reporting.
6. User Interface: Develop a user-friendly interface for hostel administrators to manage attendance records and generate reports.
7. Scalability: Design the system to handle a large number of users and accommodate future enhancements or modifications.

Overall, the system should enhance the efficiency of hostel attendance management while keeping parents informed about their child's whereabouts in real-time.

3.2 Proposed system:

The purpose of implementing a hostel attendance system using biometric fingerprint technology is to achieve several key objectives:-

1. Efficiency: The system aims to streamline the attendance tracking process by eliminating manual methods such as paper registers or card swiping. This saves time and effort for both hostel administrators and residents.
2. Accuracy: Biometric fingerprint technology provides a highly accurate method of identifying individuals, ensuring that attendance records are reliable and free from errors or falsifications.
3. Real-time Tracking: By capturing attendance based on fingerprints, the system provides real-time data on when students enter or exit the hostel premises. This helps in monitoring attendance patterns and identifying any irregularities promptly.
4. Security: Biometric data is unique to each individual and offers a high level of security compared to traditional methods like ID cards or passwords. The system ensures the privacy and integrity of biometric information.
5. Parental Communication: The system enhances communication with parents or guardians by automatically sending them notifications about their child's attendance status. This keeps parents informed and reassured about their child's activities.
6. Integration: Integrating the attendance system with the hostel management system allows for seamless data flow, centralized management of records, and easy generation of reports for analysis and decision-making.
7. Compliance: Implementing a biometric attendance system may also be a requirement for regulatory compliance or to meet institutional standards for efficient hostel management.

Overall, the purpose of the hostel attendance system using biometric fingerprint technology is to improve operational efficiency, enhance security, promote real-time monitoring, foster parental involvement, and ensure compliance with standards and regulations.

3.3 Action Plan:

Week	Task	Start Date	Finish Date
1.	Group discussion and selection of topic	01-08-2023	12-08-2023
2.	Select Title for the project. Research and read more about topic. Gathering information, group discussion and distributions of works to be don	16-08-2023	21-08-2023
3.	Gathering information about existing system	25-08-2023	10-09-2023
4.	Find drawbacks of existing syste	14-09-2023	21-09-2023
5.	Searching journals and papers related to proposed system and study them	22-09-2023	26-09-2023
6.	Make problem statement and prepare the overview of the project, designing the prototype	28-09-2023	04-10-2023
7.	Preparing UML Diagrams - Use case diagram, Activity diagram, DFD diagram, ER diagram, sequence diagram, collaboration diagram, state transition diagram	11-10-1023	18-10-2023
8.	Preparing the project report	19-10-2023	31-10-2023

Table 3.3.1: 5th Semester Action Plan

Hostel Attendance System using Biometric Finger Print Authentication

Week	Task	Start Date	Finish Date
1.	Download setup of Python for IDLE. Develop code for Login Form.	01-01-2024	03-01-2024
2.	Set up of Biometric Finger Print Machine.	04-01-2024	08-01-2024
3.	Add users and Administrator According to their Finger print Registration.	09-01-2024	12-01-2024
4.	Develop a code for Registration form	15-01-2024	19-01-2024
5.	Download setup for Mysql and Connect Mysql to Python.	22-01-2024	23-01-2024
6.	Download setup of Realtime Tracker 11.7	24-01-2024	29-01-2024
7.	Connect Laptop and Biometric machine via crossover cable to download log file of daily attendance	30-01-2024	02-02-2024
8.	Perform Testing on project	05-02-2024	08-02-2024
9.	Develop a final project by connecting all modules altogether	15-02-2024	22-02-2024

Table 3.3.2: 6th Semester Action Plan

Chapter 4

Proposed Methodology For Solving Problem

4.1 Introduction

Automated Attendance System

The attendance management is a major part of today's human resource systems. In various institutions, and academic organizations, attendance is a very essential factor which is used for different purposes. These purposes consist of record maintenance and evaluation of student's arrival. Taking of attendance is time uncontrollable and it is complicated to find out the number of students. Thus, there is a need for a system that would eliminate all of these difficulties. The Hostel management attendance system is used to verify the student's presence .

A fingerprint sensor is electronic equipment that confines a digital image of the fingerprint pattern, by some one of the fingerprint capturing technologies that might be optical, radio frequency (RF) based. The confine image is acknowledged as live scan. This live scan is then digitally method by extracting choice from the fingerprint. It is then efficiently extend into a digital uniqueness of the concerned one. It is gathered in a database, for assessment desired for authentication and access power.

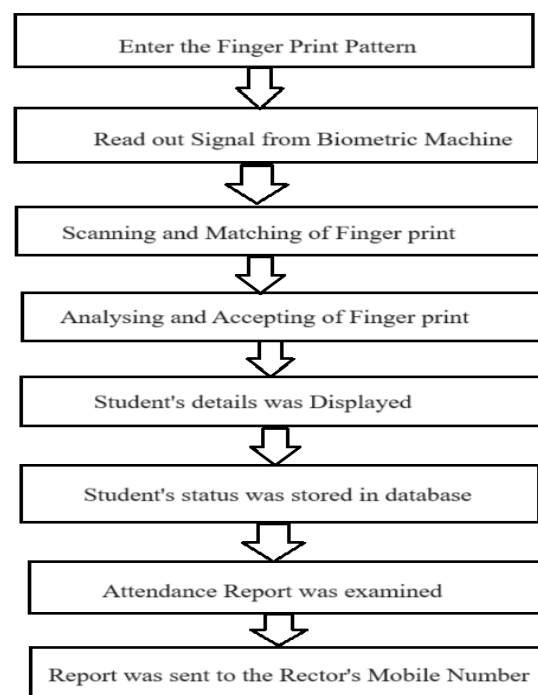
The attendance management system database consists of tables that accumulate records about the student's details that must correspond by an authorized person and has access to the system. Every record may contain the detail templates of the person's fingerprint and user name of the person. The database design system implements relational data model. That is a collection of tables in which data stored. This system having student's Name, ID Number, Register number, Course, Year, Room No, Photo, Fingerprint and parents mobile number that is stored in database. This proposed system introduces a new automatic attendance management system, which integrates fingerprint authentication into the process of attendance management for both staff and student. It is made up of two processes namely; enrolment and authentication. During enrolment, the biometrics of the user is captured and the minutiae data are extracted and stored in a database as a template for the subject along with the user's ID. The objective of the enrolment module is to admit a user using his/her ID and fingerprints into a database after feature extraction. These features form a template that is used to determine the identity of the user, formulating the process of authentication. The enrolment process is carried out by an administrator of the attendance management system. During authentication, the biometrics of the user is captured again and the extracted features are compared with the ones already existing in the database to determine a match. After a successful match, attendance is marked against the user's id used in matching the templates.

4.2 Overall Description:

Hostel Management System enhances security of the hostel students. Student's daily attendance is maintained by biometric system. Student's attendance status alerts can be sent through SMS to the parents. The absent report was very useful for the pre calculation of food preparation. This system provides an easy process for monitoring hostel student's activities by the warden. In the proposed model we can use two biometric devices; it is placed in the warden room and watchman room in main gate. The students must place their fingerprint in both of these devices. This technique increases the protection of the outlet students and used to collect the exit time and date of the students.

Along with the students attendance system their leave module and mess management also designed by this system. In mess management module we can generate mess bill preparation, mess bill collection, food preparation, stock maintenance and purchase details etc. The total number of exit students report was automatically sent to the PC located in the warden room. The module for student

leave, permission and outing also designed with this system. After getting the initial level permission from the college, the second level parent approval also processed through SMS. Parents should provide a confirmation for their wards.



After that, the gate pass was provided to the students. The gate pass will be handed over to the security within the gate by the student. After inserting a thumb in the biometric sensors, the message was sent to the parents through their registered mobile number in native language. Arrival of the student to the hostel also intimated to the parents by this same method. When the fingerprint images and the user name of a person to be enrolled are fed to the enrollment module,

a minutiae extraction algorithm is first applied to the fingerprint images and the minutiae patterns (features) are extracted. These features form a template that is used to determine the identity of the user, formulating the process of authentication. The enrolment process is carried out by an administrator of the attendance management system. The enrolment and registration phase is an administrative phase. The user fingerprint as well as other bio-data is stored for the first time into the database for registration.

The attendance management system database consists of tables that stores records, each of which corresponds to an authorized person that has access to the system. Each record may contain the minutiae templates of the person's fingerprint and user name of the person or other information such as pin no as an index to the template. The database design for the system implements relational data model which is a collections of tables in which data are stored. The database was implemented in Microsoft SQLServer database (Sql Server, 2005). SQLServer is fast and easy, it can store a very large record and requires little configuration.

The Hostel Attendance Management information regarding the details of the Hostel System is to make a separate window application for Attendance Management System. The system has been managing the hostel. The Proposed system is the developed in a way such that it can be upgraded with new computerized version of the existing system. It rectifies options easily. The whole system has been tested most of the pitfalls of the existing system and maintains thoroughly both at the development stage and in the databases of all the entities participating in the system. implementation stage. Computerization provides easy and quick access over the data. Authorization schemes like giving passwords, Future Enhancements: Once the system has been fully encoding data can be used to maintain data security.

This project is aimed at developing a system for keeping records and showing information about or in a hostel. This system will help the hostel officer to be able to manage the affairs of the hostel. This system will provide full information about a student in the hostel.

4.3 External Interface Requirements

4.3.1 Hardware Interface :

Device	-	Laptop, Biometric Fingerprint Machine T304F
Processor	-	Intel(R) Core(TM) i5-2540M CPU @ 2.60GHz 2.60 GHz
Installed RAM	-	8 GB
System Type	-	64 bit operating system, x64 bit –based processor
Windows Edition	-	Windows 10

4.3.2 Software Interface:

Frontend :- Designing GUI using Python

Backend :- MySql Database

Realtime Attendance Tracker 11.7 Software

4.3.3 Software Design:

1. Software design is a process to transform user requirements into some suitable form, which helps the programmer in software coding and implementation.

2. Software design is the first step in SDLC (Software Design Life Cycle), which moves the concentration from problem domain to solution domain.

3. It tries to specify how to fulfill the requirements mentioned in SRS.

4.3.4 Backend- Data Base (MySQL):

1. MySQL is a relational database management system based on SQL Structured Query Language.

2. The application is used for a wide range of purposes, including data warehousing, e-commerce, and logging applications.

3. The most common use for MySQL however, is for the purpose of a web database.

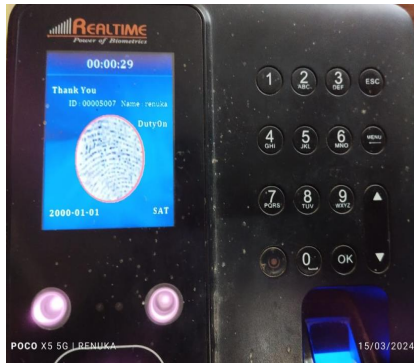
4. It can be used to store anything from a single record of information to an entire inventory of available products for an online store.

4.3.5 Frontend Python's GUI (Graphical user Interface)

Python offers multiple options for developing GUI (Graphical User Interface). Out of all the GUI methods, TKINTER is most commonly used method. It is a standard Python interface to the Tk GUI toolkit shipped with Python. Python with Tkinter outputs the fastest and easiest way to create the GUI applications. Creating a GUI using Tkinter is an easy task.

In this we use Biometric Fingerprint Machine and Realtime Attendance Tracker Software 11.7 as follows:

4.3.6 Realtime Biometric:



- Fingerprint
- TCP/IP
- Personal Identity Number
- USB Disk
- Access Control
- Camera
- Fingerprint Scanner

4.3.7 Realtime Attendance Tracker:



The significant role of Biometrics and Identity management solutions have clearly been defined since long now. Also when it comes to accuracy and reliability within Asia it is unequaled in the world today at large. Along with the business ventures, government agencies, research universities and technology business services constitute a very comprehensive and encyclopedic source of information and provide resources that greatly empower and validate business strategies to surpass their well-defined goals that presently exist Realtime Biometrics is a major performer and provider of fingerprint core technology. The fundamental principle and key ingredients behind the great success of the Biometric strategy has been mainly its two main

components (a) Its Research, (b) Its Development. The company's current successful operation is deeply rooted with a mission of accomplishing its desired aim with its knowledge base and skills.

4.4 Other Non-functional Requirements

4.4.1 Security Requirements

The Current System Security

The current system, which is Canvas, has its policy on its site page. The current system builds upon a user name and password access.

The system now has its own policy and security; however, the new feature we will add to the system will need some security requirements to the system. The new feature in the system will add some values to the current policy to maintain the security in the right way. It also provides proof of compliance.

The new policy in the system will deal with the security in many cases. The security will have more components on the system in a high control panel. The plan is to secure the outsider and insider community of misused the system (e.g. identification theft). Strong security is part of the policy's purpose.

4.4.2 User Access

Inside the community, there are students, Rector, and who are going to use the system. The main actor of the users in the entity is students. Students will use the entity everyday by scanning their fingerprints by the beginning of each class. Faculty will only use the system through Canvas, and they can access to student information. The faculty member will use the entity to control the attendance page. Faculty's job is to add, edit, update and delete any record. Furthermore, registration office staff will check on every student's identification for security purposes. They check on students for identification in person before they add, edit, update and delete any information from their fingerprint records in the system. The staff will ask students for ID for identification and print their thumb in the device if needed to make sure the person is identified.

4.4.3 Threats to the system security

This system may face many threats. Sometimes, it comes from a community insider. This could be someone who discloses the data form the database where it located, in the registration office. Another type of deception is false identification, such as a fake ID, when students present in the registration office. The system will reduce this kind of misuse because the fingerprint identification is more secure than others. Nobody can make up a fingerprint not related to them. However, Rector should check on the identification carefully before initiating any processes in the system. These records will be the official record for all students, since they begin school and until they graduate.

4.4.4 Levels of security:

- 1) Hardware: The fingerprint devices must be located in a secure location in every class. It should be behind the entrance that everyone can see the device inside and outside the class if the door is open.
- 2) The operating system: the security in this case will be in the same level of the Canvas security.
- 3) The network: it is part of the current system security.
- 4) The data management system:
 1. Students can access to their classes to the system check by fingerprint.

2. Faculty access to Canvas would be the same as we have now, and they will control the attendance page/report.
3. The purpose of the registration office is to make sure every student has the right fingerprint record and right information in the system on a consistent basis.

4.4.5 Level of access

People level:

- 1) Users (students).
- 2) Faculty

Computer level:

- 1) Hardware (fingerprint reader)
- 2) Software (the system)

4.5 System Architecture

A system architecture is the ideational model that defines the structure, behavior, technology and other views of any system. In the fingerprint system, we have the whole structure to build the system. In this architecture, we will describe the formal definition and the representation of the system. This description is a high level that can show the relationship between the components induces software, hardware and the communication between them.

The first level of this architecture is the hardware component. The hardware here is the fingerprint reader which will be connected to the system and other components. Then, it will check with security level. The security level here has database for the security purposes. Next, the architecture level will go to the process task, which has four components in the software. They are check in for every class when student scan his/her fingerprint, check on fingerprint, this is kind of the security and to check into every class. Then check into the class, which is the class check list to make sure the student in the right class. Finally, notify step, this step is to report everyone involves in the system to receive a notification.

The last two level of this architecture are the access data and the databases. There are three databases for this step which are for the classes, students, and fingerprint records. This architecture is the whole structure for the Automatic Attendance System. So, all process here will complete the communication between all the components.

4.6 System Features:

Centralized Biometric Database :

Multiple biometric systems can be managed in a single, centralized database for seamless operations.

Comprehensive Attendance Statistics :

Detailed statistics can be obtained on student attendance for a particular day, week, or year.

Individual student statistics :

Specific student status can be viewed and analyzed throughout the year.

Customizable Cut-Off Time :

The start and end time can be customized for biometric attendance registration to suit the hostel's

schedule.

No Proxy Option:

The possibility of proxy can be completely eliminated with the reliable biometric authentication system .

Approval-based SMS Notifications :

Approval based SMS notifications are sent to parents/guardians for absent students.

Customizable Design :

The system's design can be tailored to meet the specific branding and aesthetic preferences of the institution and comes with a user friendly UI

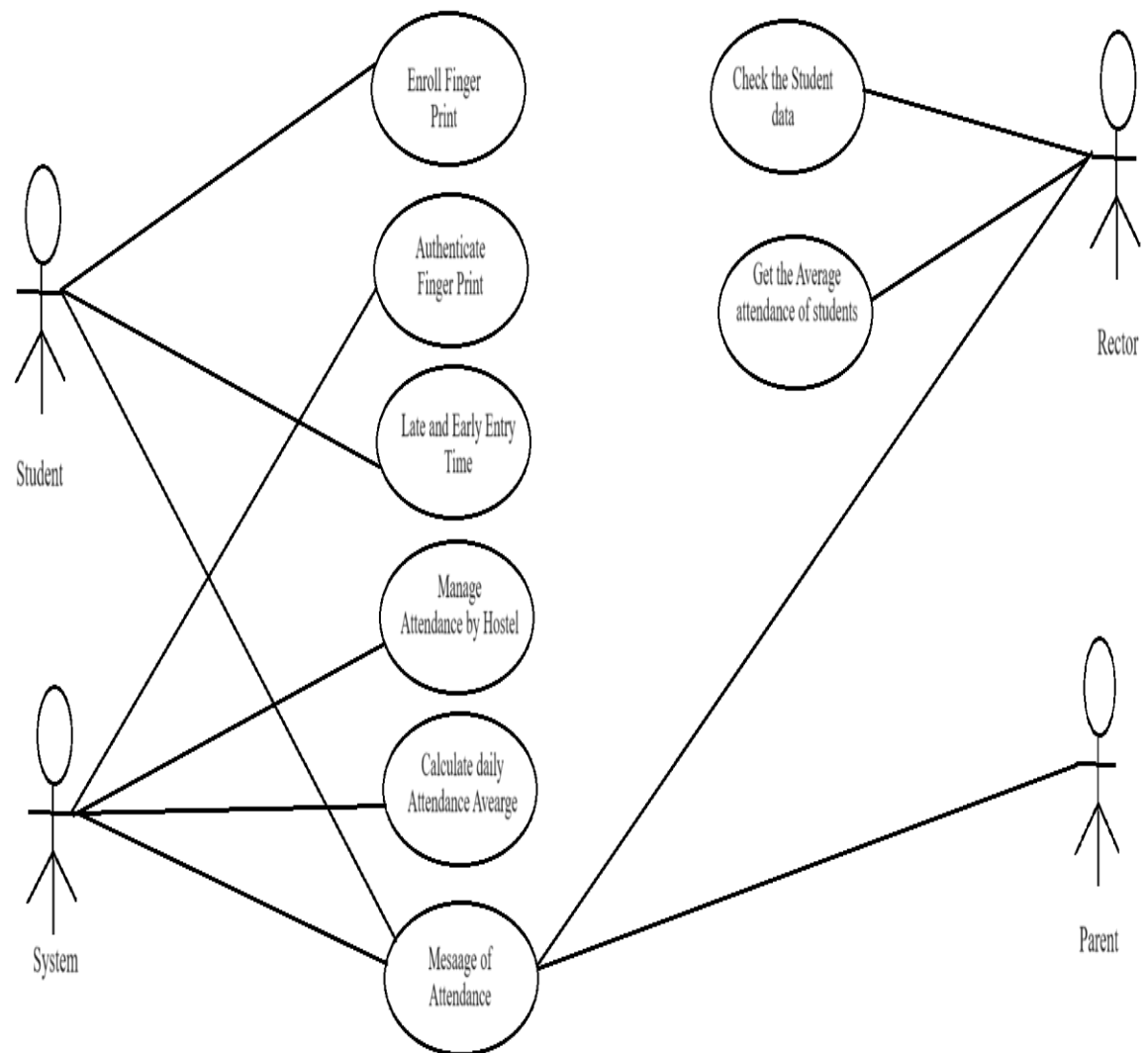
4.7 UML Diagrams:

The UML diagrams of the project are:

- Use Case Diagram
- ER Diagram
- Data Flow Diagram (DFD)

Hostel Attendance System using Biometric Finger Print Authentication

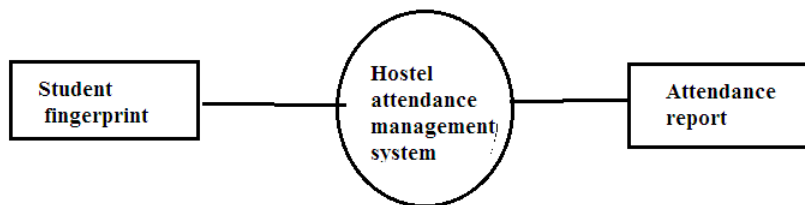
4.7.1 Use Case Diagram



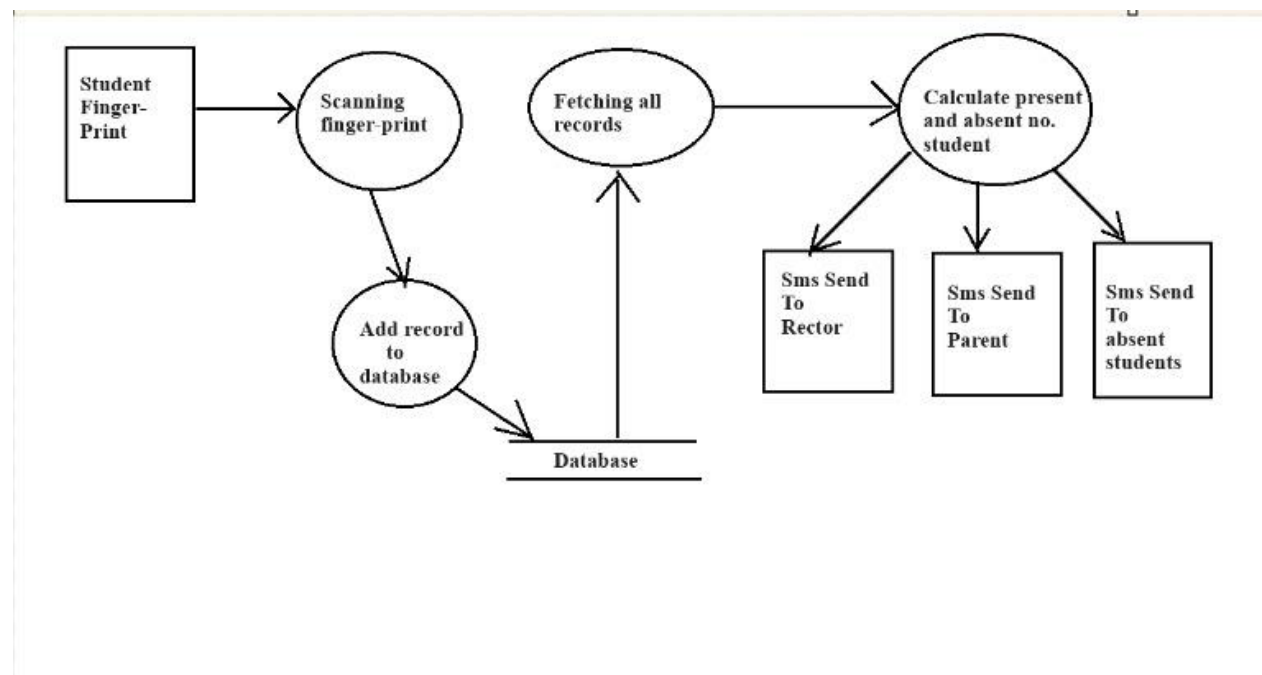
Hostel Attendance System using Biometric Finger Print Authentication

4.7.2 DFD Diagrams:

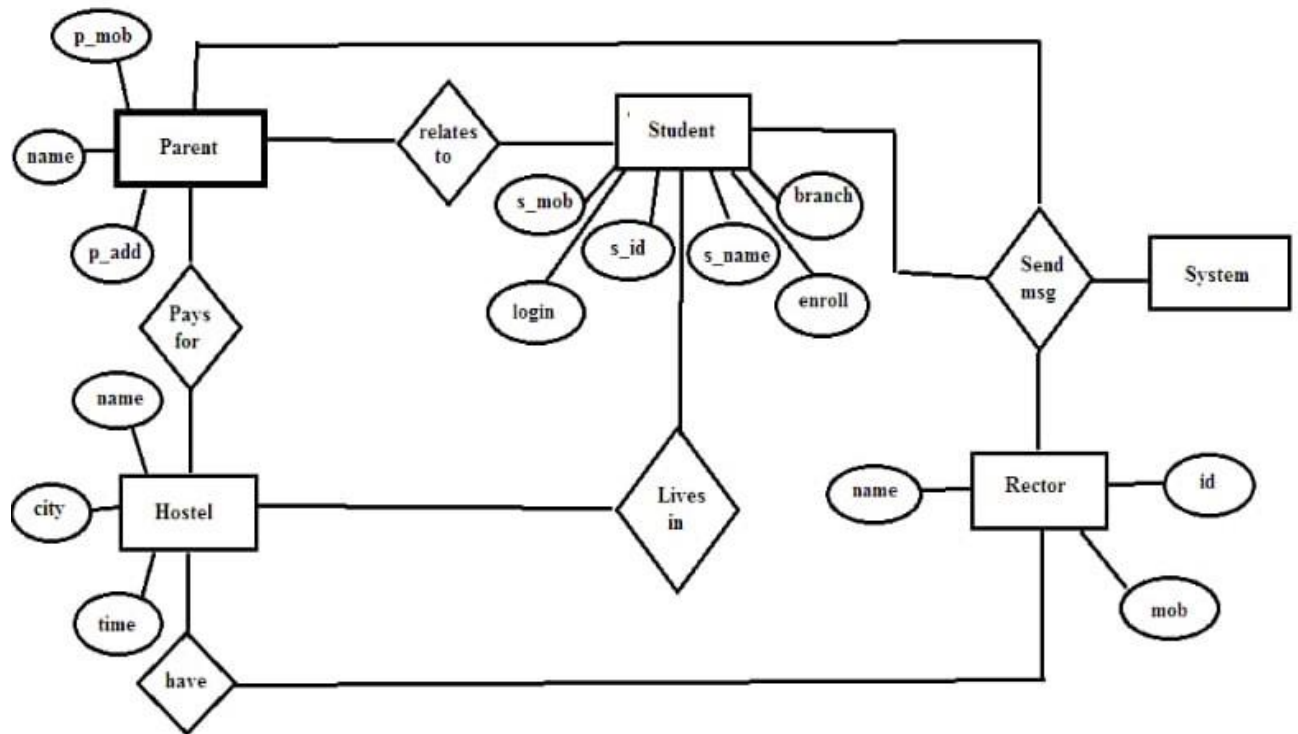
Level 0 Diagram



Level 1 Diagram



4.7.3 ER Diagram



Chapter 5

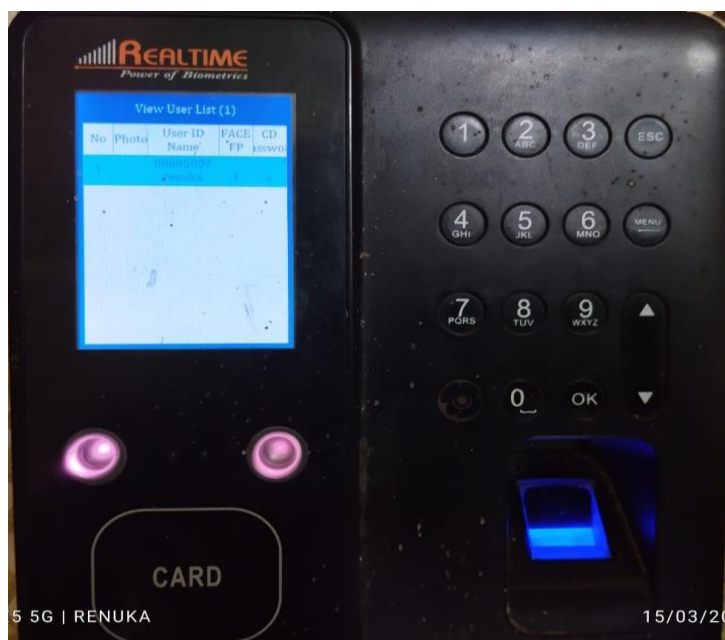
Details of designs, working and processes

5.1 Biometric Machine Setup

1. Verify Finger:



2. View User Info:

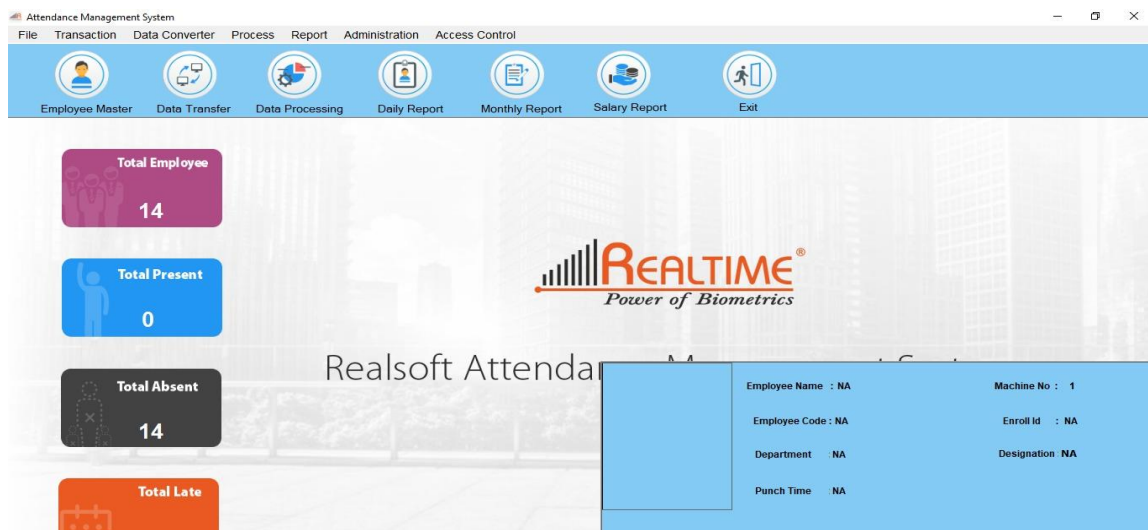


3.View Time Log:



5.2 Realtime Software login:

1.Realtime Attendance Tracker 11.7 Interface:



Hostel Attendance System using Biometric Finger Print Authentication

2.Add Sudents:

Attendance Management System

File Transaction Data Converter Process Report Administration Access Control

Employee Master

Employee List

Total Employee: 14 Total Working Employee: 14
Total Resigned Employee: 0

Select All/De-Select All

All			
☐	Unnati Kulkarni	00000006	00000001 DEMO
☐	Bhavana Patil	00000002	00000002 DEMO
☐	Leena Patil	00000008	00000003 DEMO
☐	Harshada Koli	00000007	00000004 DEMO
☐	Neha Bhamre	00000009	00000005 DEMO
☐	Gayatri Dhan...	00000010	00000006 DEMO
☐	Punam dhan...	00000011	00000009 DEMO
☐	Harshada Mo...	00000012	00000020 DEMO
☐	Renuka Bhav...	00000004	00005007 DEMO
☐	unnati	00000001	1 DEMO
☐	3	00000003	3 DEMO
☐	5	00000005	5 DEMO
☐	renuka	00005007	5007 DEMO
☐	9999	00009999	9999 DEMO

Branch Name* DEMO

Salary Mode

Shift Start Date 01-Jan-24

Second Weekly Off Days Condition

First Weekly Off Sunday

Second Weekly Off Sunday

Second WO Type None

Half Day Shift None

Shift Type* Fixed

Second Weekly off days

Rest Days will be Off

Is Auto Shift

All

001

Active Access Control

Group No

5.3 Project output

5.3.1.Login User:

GPD Hostel Attendance System

Login Form

Username admin

Password ****

Login

Login Successful

Welcome, to GPD Hostel

OK

5.3.2. Register user:

Registration form

FullName :

Mobile No:

Year : ☐ First ☐ Second ☒ Third

Enroll :

Branch :

Id :

Submit

5.3.3. Information recorded in database Table:

```

MySQL 8.3 Command Line Client
Your MySQL connection id is 31
Server version: 8.3.0 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use db1;
Database changed
mysql> select* from base1;

```

name1	mobile	Year1	enrol	Branch	id
Bhavana Patil	7385978788	Third	2100170075	Computer	00000002
Unnati Kulkarni	9921786347	Third	2100170074	Computer	00000001
Harshada Koli	9307560627	Third	2100170081	Computer	00000004
Leena Patil	9158555630	Third	2100170045	Computer	00000003
Neha Bhamare	8767462142	Third	2100170082	Computer	00000005
Gayatri Dhangar	7745280832	Third	2100170023	Computer	00000006
Punam Dhangar	9368459890	Third	2100170021	Computer	00000009
Harshada More	9975280810	Third	2100170201	Computer	00000020
Renuka Bhavsar	7745086002	Third	2100170041	Computer	00005007

```

rows in set (0.00 sec)

```

5.3.4.Data of present students:

```
Enter 1 for Registration From
Enter 2 for present student
Enter 3 for absent student
Enter 4 for Registerd all Student
2
('Renuka Bhavsar',)
('Neha Bhamare',)
('Leena Patil',)
('Bhavana Patil',)
('Gayatri Dhangar',)
('Harshada Koli',)
```

5.3.5.Data of Absent Students:

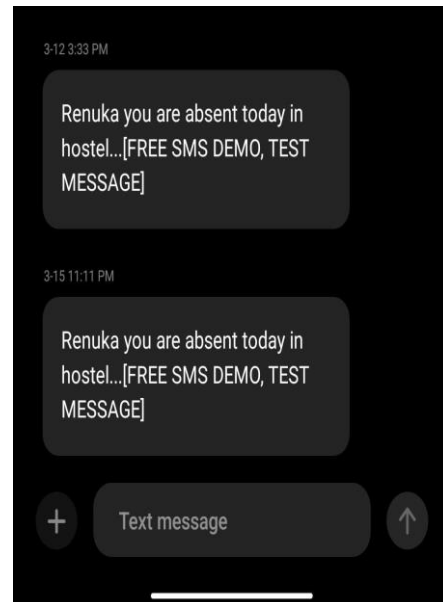
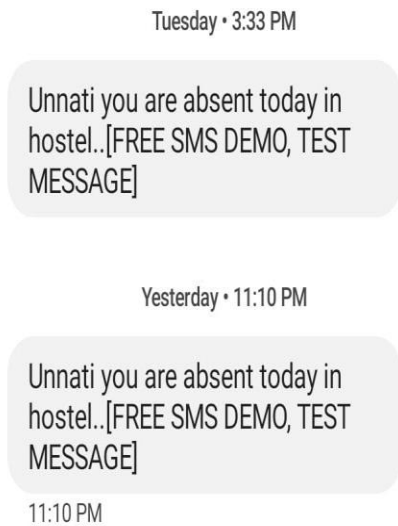
```
Enter 1 for Registration From
Enter 2 for present student
Enter 3 for absent student
Enter 4 for Registerd all Student
3
('Unnati Kulkarni', '9921786347', 'Third', '2100170074', 'Computer', '00000001')
('Punam Dhangar', '9368459890', 'Third', '2100170021', 'Computer', '00000009')
('Harshada More', '9975280810', 'Third', '2100170201', 'Computer', '00000020')
[('9921786347',), ('9368459890',), ('9975280810',)]
Hello, Rector Today There are total absent student in hostel are 3
Message sent successfully.
Message sent successfully.
Registration form is created seccussfully...
>>> |
```

5.3.6.Data of all Registered Students:

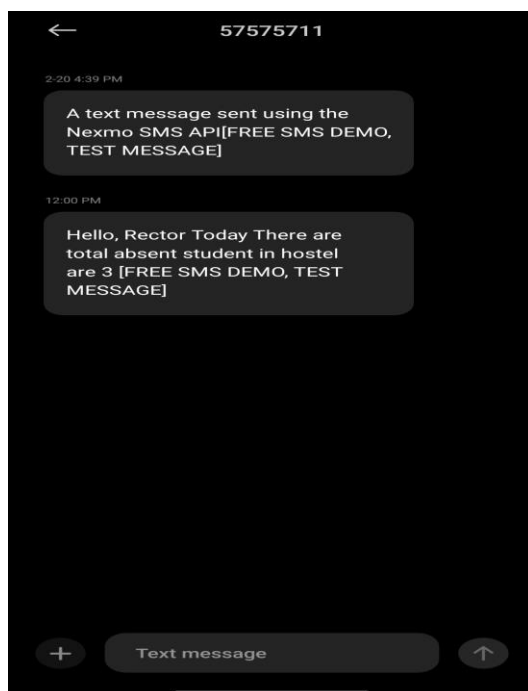
```
= RESTART: C:\Users\DELL\Desktop\project\b1o\tingoProject.py
Enter 1 for log report
Enter 2 for present student
Enter 3 for absent student
Enter 4 for Registerd all Student
4
('Bhavana Patil', '7385978788', 'Third', '2100170075', 'Computer', '00000002')
('Renuka Bhavsar', '7745086002', 'Third', '2100170075', 'Computer', '00005007\t')
('Unnati Kulkarni', '9921786347', 'Third', '2100170074', 'Computer', '00000001')
('Harshada Koli', '9307560627', 'Third', '2100170081', 'Computer', '00000004')
('Leena Patil', '9158555630', 'Third', '2100170045', 'Computer', '00000003')
('Neha Bhamare', '8767462142', 'Third', '2100170082', 'Computer', '00000005')
('Gayatri Dhangar', '7745280832', 'Third', '2100170023', 'Computer', '00000006')
('Punam Dhangar', '9368459890', 'Third', '2100170021', 'Computer', '00000009')
('Harshada More', '9975280810', 'Third', '2100170201', 'Computer', '00000020')
Registration form is created seccussfully...
```

5.4 Messages Received By Parents, Student and Rector:

1. Message Received by absent students and Parents :



2.Message Received by Rector



Chapter 6

Results and Applications:

6.1 Test Cases:

Test case Id	Condition to be Tested	Expected Result	Actual Result	Status
TC_1	Check whether the Username and Password are correct or not	Login Successfully message should be displayed	Login Successfully message is display	Pass
TC_2	Check whether Hosteller name contains other than characters	Name should contain only alphabets	Name accepts only alphabets in the name Field	Pass
TC_3	Check whether the mobile number contains only digis	Mobile number field should Accepts only numeric data	Mobile number field Accepts only numeric data	pass
TC_4	Check whether contact number contains exactly 10 digits	Contact number should not accepts more than 10 digits	Contact number not accepts more than 10 digits	Pass
TC_5	Check Whether Finger print Machine login only for Registered Thumb rather than any other thumb	It should not accept any other finger as an input	It not accept any other finger as an input . Show an error	Pass
TC_6	Check Whether that it display correct total Number of Present students	It should display Correct Calculation of total number of present students	It displays Correct Calculation of total number of present students	Pass
TC_7	Check Whether that it display correct total Number of Absent students	It should display Correct Calculation of total number of Absent students	It displays Correct Calculation of total number of Absent students	Pass

Hostel Attendance System using Biometric Finger Print Authentication

TC_8	Check Whether It display Correct number of Total Students	It should display Correct total number of students Registered in Hostel	It displays Correct total number of students Registered in Hostel	Pass
TC_9	Check whether data fill into the form correctly recorded in database table	Data should be correctly recorded in database table	Data is correctly recorded in database table	Pass
TC_10	Check whether All absent students get message from hostel	All absent Students should get message from hostel	All absent Students get message from hostel	Pass
TC_11	Check whether the Message to successfully deliver to the parents of Absent Students	The Message should successfully delivered to the parents of absent students	The Message is successfully delivered to the parents of absent students	Pass
TC_12	Check whether the correct calculation of number absent students are delivered to the Rector of hostel through Message successfully	Rector should be received message of number of absent students	Rector is received message of number of absent students	Pass

6.2 : Applications:

- 1.Fingerprint biometrics is used to both authenticate (match a person's biometric template) and identify (determine the identification of a person).
- 2.A person's unique physical characteristics are called biometrics, which can be used to validate and verify identity.
- 3.Fingerprint biometrics is some of the most commonly used types of biometrics for authentication and identification purposes, but facial recognition, iris scanning, and DNA are all forms of biometric identifiers as well. Fingerprint biometrics is widely distributed, cost-effective, easy to use, effective, hard to fake, and convenient, making them a secure and useful tool for authentication and identity verification.
- 4.It is also used as College attendance system.
- 5.By using this system parents cannot worry about their child that they are safe in hostel and get their daily absent and present status.

6.3 : Key Benifits

As the world continues to become more technologically advanced, school hostels are not left behind. Hostel student management software is an essential tool for any school hostel. It helps manage student records, track attendance, and communicate with parents effectively

1.Efficient Communication:

Effective communication is critical in any institution, and school hostels are no exception. Hostel student management software allows hostel administrators to send important updates, announcements, and alerts to parents and students in real-time. This feature ensures that everyone is informed promptly and can respond accordingly.

2.Improved Record-Keeping:

Hostel student management software allows administrators to maintain an accurate record of student data, including personal information, academic records, attendance, and discipline records. With this information, the hostel management can monitor student performance and behavior and take appropriate action where necessary.

3.Easy Attendance Tracking:

Attendance tracking is an essential part of hostel management. Hostel student management software provides an easy way to track attendance, reducing the likelihood of errors and increasing the accuracy of the data. The software can also generate reports that help administrators track trends and identify patterns.

4.Enhanced Security:

Hostel student management software helps enhance security in the school hostel. The software can track student movements, monitor visitors, and restrict access to certain areas of the hostel, ensuring that students are safe and secure at all times.

5. Manage the Number of Students

One of the primary tasks of the hostel warden at the start of each year is to supervise student admissions while keeping in mind the maximum number of students the hostel can handle. Wardens can record information regarding the hostel, such as new admissions or if any student is leaving the hostel. With the help of this feature, wardens will have complete control over the number of students present in the hostel. They can easily track the students who have enrolled in the hostel or delete the record of those students who have left the hostel.

6.Reduces The Manual Work Of The Administrative Staff

Managing the daily attendance of students at a hostel - their incoming and outgoing times - can be a time-consuming task, particularly in medium and large-sized hostels. In this scenario, the hostel attendance management system makes it easier to manage attendance. It also decreases the possibility of human error.

7. Hostel attendance monitoring:

systems can help schools protect their students. Students often feel safer staying in hostels because many hostels require that you have a hostelling card to get a room. Hostel cards often include information about the student, which is only updated when the student checks in at night or during the day to use their room. When the school knows who is on board with the student and their activities, they know who their students are and can help keep them safe from sexual assault. Hostel attendance monitoring systems are easy to use and make you feel safe when your students live on campus.

8. Time Management:

They can help you avoid liability issues by making sure that students are present during their time at the hostel.

6.4 Limitations of A Biometric Attendance System:

1. False Positives

False positives happen when a biometric system incorrectly accepts an unauthorized user. This can happen due to stolen biometric data used to spoof the system. For example, a fingerprint lifted off a surface could beat a fingerprint scanner.

Facial recognition is also prone to false positives due to lookalike individuals. The consequences of false positives include data or facility access given to imposters. Setting stringent matching thresholds can reduce this risk but increases rejections.

2. Scanner Compatibility Issues

Incompatibility between users' biometrics and scanners can cause enrollment and authentication failures. For fingerprint systems, factors like dry skin, scars, temporary cuts, etc., affect imaging. Iris scanners can fail.

3. High Costs

Implementing an enterprise-grade biometric system requires significant upfront investment. Costs arise from specialized scanners, sensing devices, data storage, and transmission systems. Ongoing expenses like maintenance, upgrades, and user training add up too. As biometric accuracy relies heavily on scanner quality, cutting corners leads to subpar performance. However, costs are decreasing over time as technologies mature and become mainstream.

4. Enrollment Difficulties

Users may experience difficulty providing biometric samples during enrollment. Data capture fails more often for fingerprints with age-related skin changes, medical conditions, injuries, etc. Mobility and health issues can inhibit positioning for face/iris scans, too. Disabilities like amputations or visual impairment limit certain modalities. Guiding users through ideal placement and conditions improves success. Expanding enrollment times helps capture accurate data.

5. Data Security Threats

As biometric traits are unique personal identifiers, a data breach poses serious privacy risks. Hacking or compromising the biometric database allows identity theft. Spoofing scanners with raw data lifts security. Biometric data should undergo encryption using irreversible cryptographic hashes for storing and transmission. Tamper-proof hardware secures the data lifecycle within systems.

Chapter 7

Conclusion and Future scope

7.1 Conclusion:

Hostel Management System is very helpful for hostel attendance system, student's leave, etc.; this hostel management software package is designed for people who wish to manage numerous activities within the hostel.

This proposal will reduce manpower, and investment of the management. By this system we can easily approach and pass the information to the parents

Biometric technology may be a reliable tool for authentication.

The existing systems can further be improved or combined which helps in making the system more users friendly.

To conclude the description about the project, the project, developed using PHP with MySQL is based on the requirement specification of the user and the analysis of the existing system, with flexibility for future enhancement.

Hostel Management System is very useful for hostel allotment and mess fee calculation. This hostel management software is designed for people who want to manage various activities in the hostel. For the past few years the numbers of educational institutions are increasing rapidly. Thereby the numbers of hostels are also increasing for the accommodation of the students studying in this institution. And hence there is a lot of strain on the person who are running the hostel and software's are not usually used in this context.

This particular project deals with the problems on managing a hostel and avoids the problems which occur when carried manually.

Identification of the drawbacks of the existing system leads to the designing of computerized system that will be compatible to the existing system with the system which is more user friendly and more GUI oriented.

The system successfully took the attendance both at lectures and examinations. The prototype successfully captured new fingerprints to be stored in the database; scanned fingerprints placed on the device sensor and compared them against those stored in the database successfully. The performance of the system was acceptable and would be considered for full implementation especially because of its short execution time and reports generation.

The objective of the enrolment module is to admit a user using his/her ID and fingerprints into a database after feature extraction. These features form a template that is used to determine the identity of the user, formulating the process of authentication. The enrolment process is carried out by an administrator of the attendance management system. During authentication, the biometrics of the user is captured again and the extracted features are compared with the ones already existing in the database to determine a match. After a successful match, attendance is marked against the user's id used in matching the templates.

Biometric attendance systems offer a glimpse into the future of work, presenting several potential benefits for efficiency, flexibility, and performance management. However, careful consideration of the associated challenges, particularly regarding privacy, cost-effectiveness, inclusivity, and potential biases, is essential for responsible and ethical implementation. As

technology continues to evolve and regulations adapt, navigating these challenges will be crucial for businesses to harness the potential of biometric systems while fostering a trustworthy and inclusive work environment.

7.2 Future Scope:

1.We can also build this project by using cloud computing by means our data is stored on cloud rather than store it manually and everyone should access it. And by using cloud computing our storage is also saved. This project become more flexible by using cloud computing.

2.On this level our project is perfectly work on 10-11 girls in hostel. We can make several improvement in that project that it will be work properly on the data of 150-170 girls. And we can also we it for the purpose of boy's hostel.

3.In this project we also add the module of location by this we can track the location of registered mobile number and if the location is out of city of college the we will inform their parents about their current location. So that the parents have no need to worry much about their children.

4.By using this attendance system we will design the aatendance system for college for each year students.If they are absent in classroom the message should be send to their parents and particular faculty also. And it will be the better improvement for college because now a days students are not attending their lecture property and therefore the will be loss in their studies as well.

5.Biometric verification is gaining a lot of popularity among the public security systems as well as in the commercial market. In our daily life, we witness the use of biometric in so many places such as the digital attendance system at offices, security checkpoints at airports, wearable tech gadgets retrieving our biological information, and even our national ID cards aka Aadhaar cards are created using biometrics technology. In fact, this national ID program holds the largest biometric database in the world.

6.In order to undertake biometric authentication, there are several methods of collecting the data. Each of these methods has its own importance whether performed separately or combined with other forms of data collection. Interestingly, some forms of biometric authentication can be performed even without the knowledge of the subject under surveillance

7.Fingerprint identification refers to the procedure of comparing the pattern of ridges, furrows and minutiae points of the fingers. It has been reported that after comparing the database of fingerprints collected in more than 140 years, no two fingerprints were found to be the same, not even for the identical twins.

8.All the law enforcement agencies across all levels of Federal Government use biometric technology to identify the suspects. The most traditional method used for this purpose is the Fingerprint Recognition, but nowadays facial and vein pattern recognition methods are also being used in the market.

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- 12.<https://chat.openai.com/>
- 13.<https://gemini.google.com/>
- 14.<https://www.vonage.com/>
- 15.<https://www.google.com/>
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- 17.<https://www.ResearchGate.com/>
- 18.<https://www.python.org/doc/>